Probeklausur: Microwave Engineering (MWE)

Master-Semester – Vollständige Klausur-Referenz & Formelsammlung

Inhaltsverzeichnis

1 Important General Information / Wichtige allgemeine Hinweise **2**
1.1 Punkteverteilung / Credit Overview . . . 2

2 Part 1: Paper & Broadband GaN/InP MMIC Design **2**
2.1 Task 1 – 40.0 Credits . . . 2

3 Part 2: General Questions **3**
3.1 Task 2: Power Amplifiers, Nonlinearity & Dynamic Range – 20.0 Credits . . . 3
3.2 Task 3: Two-Port Stability, Smith-Chart Matching & Planar Transmission Lines – 20.0 Credits . . . 4
3.3 Task 4: Low Noise Amplifiers (LNAs), Cascaded Noise & Simultaneous Matching – 20.0 Credits . . . 5

4 Lösungen zu Part 1: Paper & Broadband GaN/InP MMIC Design (Task 1 – 40.0 Credits) **6**
4.1 Lösung 1.1 (2.5 Credits) . . . 6
4.2 Lösung 1.2 (1.5 Credits) . . . 6
4.3 Lösung 1.3 (3.0 Credits) . . . 7
4.4 Lösung 1.4 (6.5 Credits) . . . 7
4.5 Lösung 1.5 (6.0 Credits) . . . 7
4.6 Lösung 1.6 (2.0 Credits) . . . 8
4.7 Lösung 1.7 (2.0 Credits) . . . 8
4.8 Lösung 1.8 (4.5 Credits) . . . 8
4.9 Lösung 1.9 (3.0 Credits) . . . 9
4.10 Lösung 1.10 (3.0 Credits) . . . 9
4.11 Lösung 1.11 (4.0 Credits) . . . 9
4.12 Lösung 1.12 (2.0 Credits) . . . 9

5 Lösungen zu Part 2: General Questions (Tasks 2, 3, 4 – 60.0 Credits) **9**
5.1 Lösungen zu Task 2: Power Amplifiers, Nonlinearity & Dynamic Range (20.0 Credits) . . . 9
5.2 Lösungen zu Task 3: Two-Port Stability, Smith-Chart Matching & Planar Transmission Lines (20.0 Credits) . . . 11
5.3 Lösungen zu Task 4: Low Noise Amplifiers (LNAs), Cascaded Noise & Simultaneous Matching (20.0 Credits) . . . 13

Universität Stuttgart – Institut für Robuste Leistungshalbleitersysteme (ILH) Prüfer: Prof. Dr.-Ing. Ingmar Kallfass **Bearbeitungszeit:** 120 Minuten **Gesamtpunktzahl:** 100 Credits

1 Important General Information / Wichtige allgemeine Hinweise

1. Enter your name and your matriculation number on this and on each following sheet with answers to the exam questions. *(Tragen Sie Ihren Namen und Ihre Matrikelnummer auf diesem und jedem folgenden Lösungsbogen ein.)*
2. Do not use red color or lead pencils. *(Verwenden Sie keine rote Farbe und keine Bleistifte.)*
3. Place your student identity card clearly visible on the table. *(Legen Sie Ihren Studierendenausweis gut sichtbar auf den Tisch.)*
4. You have **120 minutes** to complete the tasks. *(Sie haben 120 Minuten Zeit zur Bearbeitung der Aufgaben.)*
5. Before you start, check that your copy of the exam contains all pages and the scientific paper *"Evolution of the Nonuniform Distributed Power Amplifier"* (C. F. Campbell, IEEE Microwave Magazine).
6. **Important Rule for Data Sheet / Paper Values:** For each value you take from the data sheet or paper, clearly mark them on the solution sheet (*"Value from page X of data sheet/paper"*) and on the data sheet/paper (*circle the value and write "Value for task X.Y"*). **In calculation tasks, always state the correct physical unit (penalties of** $-0{,}25$ **credits apply for missing or wrong units!).**

1.1 Punkteverteilung / Credit Overview

Teil / Part	**Part 1 (Paper & Broadband MMICs)**	**Part 2 (General Questions)**	**Part 2 (General Questions)**	**Part 2 (General Questions)**	**Sum**
Task / Aufgabe	**Task 1**	**Task 2 (PA & Nonlinear)**	**Task 3 (Stability, Smith Chart & TRLs)**	**Task 4 (LNA & Noise)**	**Total**
Maximum Credits	**40.0**	**20.0**	**20.0**	**20.0**	**100.0**
Achieved Credits					

Teil I: Klausuraufgaben (ohne Lösungen)

2 Part 1: Paper & Broadband GaN/InP MMIC Design

Reference Paper: Charles F. Campbell, *Ëvolution of the Nonuniform Distributed Power Amplifier"*, IEEE Microwave Magazine, 2019 (and Lecture Chapters 2, 3, 5, 6).

2.1 Task 1 – 40.0 Credits

- **1.1 (2.5 Credits)** Compare Gallium Nitride (GaN) HEMTs and Indium Phosphide (InP) HEMTs with respect to breakdown voltage, cutoff frequencies ($f_T, f_{\max}$), and their primary application in millimeter-wave frontends. State the fundamental physical relation (**Johnson Figure of Merit**) that links breakdown voltage and transit frequency within a given semiconductor material.
- **1.2 (1.5 Credits)** Why is Silicon Carbide (SiC) preferred over Sapphire or bulk Gallium Arsenide (GaAs) as a substrate for the epitaxial growth of GaN power transistors, and why does high-voltage operation of GaN transistors pose a severe challenge for wideband reactive matching?

- **1.3 (3.0 Credits)** Draw the simplified large-signal **Norton equivalent circuit** of the output of a microwave power FET (the input can be considered ideal with gate-source voltage v_{gs}) and specify how its equivalent circuit elements ($R_p = 1/G_p$ and C_p) are experimentally determined in practice.
- **1.4 (6.5 Credits)** Assume a GaN power transistor with a maximum drain current capability of $I_{\max} = 4\,\mathrm{A}$, a knee voltage of $V_{\mathrm{knee}} = 200\,\mathrm{mV}$, and a drain supply voltage of $V_{\mathrm{D,supply}} = 28\,\mathrm{V}$. The parallel output capacitance of the Norton equivalent circuit is $C_p = 1{,}6\,\mathrm{pF}$ and the load-line fitting parameter is $\beta = 1{,}5$. Calculate the maximum theoretically achievable 15 **dB return-loss matching bandwidth** ($\mathrm{RL_{dB}} = 15\,\mathrm{dB}$) according to the Bode-Fano criterion derived in the paper:

$$\mathrm{BW} < \frac{4{,}343 \cdot I_{\max}}{\mathrm{RL_{dB}} \cdot C_p \cdot \beta \cdot (V_{\mathrm{D,supply}} - V_{\mathrm{knee}})}$$

- **1.5 (6.0 Credits)** The maximum current of $I_{\max} = 4\,\mathrm{A}$ of the GaN device from Task 1.4 is obtained for a total gate width (TGW) of $W_{\mathrm{tot}} = 5\,\mathrm{mm}$ (i.e., a normalized current capability of $I_{\max,\mathrm{norm}} = 800\,\mathrm{mA/mm} = 0{,}8\,\mathrm{A/mm}$).
  1. Calculate the **normalized real part** $R_{p,\mathrm{norm}}$ (in $\Omega \cdot \mathrm{mm}$) and the **absolute resistance** R_p (in Ω) of the Norton equivalent circuit for this transistor.
  2. Calculate the maximum linear Class-A RF output power $P_{\mathrm{out,max}}$ (in W and in dBm) according to Cripps' load-line theory for $\beta = 1$.
- **1.6 (2.0 Credits)** Name two distinct design or operating options to increase the theoretically achievable matching bandwidth if the GaN transistor from Task 1.4 is used in a classical reactively matched common-source amplifier configuration. What is the respective trade-off of each option?
- **1.7 (2.0 Credits)** Explain the fundamental concept of **distributed amplification (Travelling-Wave Amplifier, TWA)** to overcome the Bode-Fano bandwidth limitation, and name the crucial passive circuit structures formed by embedding the transistor capacitances into inductive interconnect lines.
- **1.8 (4.5 Credits)** In a uniform n-stage Travelling-Wave Amplifier (TWA), each transistor stage has an effective input capacitance of $C_{\mathrm{in}} = 120\,\mathrm{fF}$ and an output capacitance of $C_{\mathrm{out}} = 120\,\mathrm{fF}$ (after capacitive equalization). The interconnecting transmission line sections of length $l_g = l_d = 200\,\mu\mathrm{m}$ contribute a line inductance of $L_{\mathrm{sec}} = L' \cdot l = 350\,\mathrm{pH}$ and a line capacitance of $C_{\mathrm{sec}} = C' \cdot l = 20\,\mathrm{fF}$ per section.
  1. Calculate the effective characteristic impedance $Z_{L,g} = Z_{L,d}$ of the artificial gate and drain transmission lines.
  2. Calculate the **Bragg cutoff frequency** $f_c = \frac{1}{\pi\sqrt{L_{\mathrm{sec}}(C_{\mathrm{sec}}+C_{\mathrm{in}})}}$ of the artificial transmission line.
  3. State the phase synchronization condition between the gate and drain lines required for constructive wave summation at the output.
- **1.9 (3.0 Credits)** For a standalone FET, is the input capacitance C_{in} or the output capacitance C_{out} usually larger? What consequence does this have for a distributed amplifier, and how is this problem solved on the gate line?
- **1.10 (3.0 Credits)** Give **three reasons** why a **uniform** distributed amplifier performs well as a low-noise or small-signal broadband amplifier, but exhibits poor efficiency and reduced output power when operated as a Power Amplifier (PA).
- **1.11 (4.0 Credits)** Which circuit topology adapts the distributed amplifier to the requirements of a high-efficiency wideband PA? Describe exactly how the **series gate capacitors** ($C_{g,k}$) and the **characteristic impedance of the drain transmission line** ($Z_{0,d,k}$) must be tapered from the input (stage 1) towards the output (stage n), and state which geometrical layout parameter of a microstrip line controls its characteristic impedance.
- **1.12 (2.0 Credits)** Which two transmission-line transformer topologies are discussed in the Campbell paper regarding bandwidth limitations at the low-impedance output of an NDPA, and which alternative impedance transformer is actually used in the final MMIC layout (Fig. 13 of the paper)?

---

3 Part 2: General Questions

3.1 Task 2: Power Amplifiers, Nonlinearity & Dynamic Range – 20.0 Credits

A millimeter-wave solid-state power amplifier (SSPA) for a 77 GHz transceiver is characterized in a $50\,\Omega$ measurement setup.

- At low input power levels, the amplifier exhibits a linear small-signal gain of $G_0 = 16\,\mathrm{dB}$.

- The input-related 1 dB compression point is measured as $\mathrm{ICP}_{1\mathrm{dB}} = +6\,\mathrm{dBm}$.
- At the 1 dB compression point, the amplifier draws a DC current of $I_{\mathrm{DC}} = 120\,\mathrm{mA}$ from a $V_{DD} = 5\,\mathrm{V}$ supply.
- The saturated output power of the amplifier is $P_{\mathrm{sat}} = +23\,\mathrm{dBm}$.

Questions for Task 2:

- **2.1 (2.5 Credits)** Sketch the qualitative curves of output power P_{out} [dBm] and gain G [dB] versus input power P_{in} [dBm] for a one-tone power sweep. Clearly mark and label the points $P_1 = \mathrm{ICP}_{1\mathrm{dB}}$, $P_2 = \mathrm{OCP}_{1\mathrm{dB}}$, $P_3 = P_{\mathrm{sat}}$, small-signal gain G_0, and compressed gain $G_{1\mathrm{dB}}$.
- **2.2 (4.5 Credits)** Calculate for operation exactly at the 1 **dB compression point**:
  1. The output-related 1 dB compression point $\mathrm{OCP}_{1\mathrm{dB}}$ in dBm and in mW,
  2. The DC power consumption P_{DC} in mW,
  3. The **drain efficiency** η_D (in %) and the **Power-Added Efficiency (PAE)** (in %).
- **2.3 (2.0 Credits)** Which of the two power metrics ($\mathrm{OCP}_{1\mathrm{dB}}$ vs. P_{sat}) is the primary design target for an **OFDM/QAM communication transmitter** and which one for an **FMCW automotive radar transmitter**? Briefly justify both answers.
- **2.4 (2.0 Credits)** In a nonlinear Taylor series expansion of the drain current around the quiescent point I_0,

$$I(t) = I_0 + a_1 v_{\mathrm{in}}(t) + a_2 v_{\mathrm{in}}^2(t) + a_3 v_{\mathrm{in}}^3(t),$$

  which polynomial term (a_1, a_2, or a_3) causes the **DC operating point shift (Self-Biasing)** under large-signal single-tone excitation $v_{\mathrm{in}}(t) = v_1 \sin(\omega_0 t)$, and which term causes **gain compression** at the fundamental frequency ω_0? How do their amplitudes scale with v_1?
- **2.5 (3.0 Credits)** Now a **two-tone test** with two closely spaced equal-amplitude tones at $f_1 = 77{,}00\,\mathrm{GHz}$ and $f_2 = 77{,}01\,\mathrm{GHz}$ is performed.
  1. At which exact frequencies do the third-order intermodulation (IM_3) products appear, and why are they much more harmful for in-band amplification than the second- and third-order harmonics ($2f_1, 3f_1$)?
  2. With what slope (in dB/dB) does the IM_3 output power increase with input power under weak nonlinearity?
- **2.6 (4.0 Credits)** In the two-tone measurement of the amplifier ($G_0 = 16\,\mathrm{dB}$, $\mathrm{NF} = 7\,\mathrm{dB}$, measurement bandwidth $B = 100\,\mathrm{MHz}$, thermal noise density $kT_0 = -174\,\mathrm{dBm/Hz}$), at an output power of $P_{\mathrm{out}} = +10\,\mathrm{dBm}$ per tone, the IM_3 products are suppressed by $\Delta P = 40\,\mathrm{dBc}$ below the fundamental tones.
  1. Calculate the output-referred third-order intercept point $\mathbf{OIP}_3$ and the input-referred $\mathbf{IIP}_3$ (in dBm).
  2. Calculate the minimum detectable signal at the input ($\mathbf{MDS_{in}}$) and output ($\mathbf{MDS_{out}}$) in dBm, as well as the **Spurious-Free Dynamic Range (SFDR)** in dB.
- **2.7 (2.0 Credits)** State the conduction angle θ_c and the maximum theoretical drain efficiency $\eta_{\max}$ for **Class-A** and **Class-B** operation. Why do parallel multi-transistor PA stages require **odd-mode suppression resistors** between adjacent transistor branches?

3.2 Task 3: Two-Port Stability, Smith-Chart Matching & Planar Transmission Lines – 20.0 Credits

A millimeter-wave HEMT transistor in common-source configuration is characterized by its $50\,\Omega$ S-parameters at $f_0 = 60\,\mathrm{GHz}$:

$$S_{11} = 0{,}6 \cdot e^{-j90^\circ} = -j0{,}6, \qquad S_{12} = 0{,}1 \cdot e^{j0^\circ} = 0{,}1, \qquad S_{21} = 2{,}5 \cdot e^{j90^\circ} = j2{,}5, \qquad S_{22} = 0{,}5 \cdot e^{-j90^\circ} = -j0{,}5$$

Questions for Task 3:

- **3.1 (5.0 Credits)** Evaluate the stability and maximum gain of this transistor at $f_0 = 60\,\mathrm{GHz}$:
  1. Calculate the determinant $\Delta = S_{11}S_{22} - S_{12}S_{21}$, its magnitude $|\Delta|$, and the **Rollett stability factor** k.
  2. Is the transistor **unconditionally stable** at 60 GHz? Justify your answer using all necessary conditions.

3. Calculate the **Maximum Stable Gain (MSG)** in dB and, if applicable, the **Maximum Available Gain (MAG)** in dB.

- **3.2 (4.0 Credits)** Suppose the output impedance of an amplifier stage is purely real and equal to $Z_{\text{out}} = 12{,}5\,\Omega$ (i.e. normalized impedance $z_{\text{out}} = 0{,}25$ in a $Z_0 = 50\,\Omega$ system) and needs to be matched to the load impedance $Z_{\text{goal}} = 50\,\Omega$ ($z_{\text{goal}} = 1{,}0$) at $f_0 = 60\,\text{GHz}$ using a lossless **2-element LC L-network**:

  1. Draw the **two possible LC matching network topologies** (Topology A: Low-Pass L-network; Topology B: High-Pass L-network) including the terminations Z_{out} and Z_{goal}, and state for each element whether it is connected in **series** or in **shunt (parallel)**.
  2. Describe the exact transformation paths of both topologies in the Smith chart (specifying the circles and clockwise / counter-clockwise directions).

- **3.3 (3.0 Credits)** How do the matching bandwidths of Topology A and Topology B from Task 3.2 compare? Which theoretical criterion governs this bandwidth, and how is it represented graphically in the Smith chart?

- **3.4 (4.0 Credits)** At millimeter-wave frequencies ($f_0 = 60\,\text{GHz}$, effective permittivity of the substrate $\varepsilon_{\text{eff}} = 7{,}84$), lumped inductors and capacitors are replaced by planar transmission line elements:

  1. Which transmission line elements replace a **shunt capacitor**, a **shunt inductor**, and a **series inductor**, respectively?
  2. Instead of a 2-element LC network, how can the real impedance $Z_{\text{out}} = 12{,}5\,\Omega$ be matched to $Z_{\text{goal}} = 50\,\Omega$ using **only a single distributed element**? Calculate its required **characteristic impedance** Z_T (in Ω) and its **physical length** l (in μm) at $f_0 = 60\,\text{GHz}$ ($c_0 = 3 \cdot 10^8\,\text{m/s}$).

- **3.5 (4.0 Credits)** Compare **Microstrip Lines (MSL)** and **Coplanar Waveguides (CPW)** for monolithic microwave integrated circuits (MMICs):

  1. Give two advantages and two disadvantages of CPW compared to MSL.
  2. Which parasitic propagation mode occurs on a CPW at discontinuities if the two ground planes are at different potentials, and which technological feature is required to suppress it?
  3. How can parasitic **substrate modes** above 100 GHz be suppressed, and how is the resulting waveguide structure called?

3.3 Task 4: Low Noise Amplifiers (LNAs), Cascaded Noise & Simultaneous Matching – 20.0 Credits

A millimeter-wave receiver frontend operating at $f_0 = 60\,\text{GHz}$ consists of a cascade of three stages connected to an antenna:

- **Stage 1 (Passive bandpass filter & feed line between antenna and LNA):** Insertion loss $L_1 = 2\,\text{dB}$ (in thermal equilibrium at $T_0 = 290\,\text{K}$).
- **Stage 2 (Low Noise Amplifier – LNA):** Gain $G_2 = 20\,\text{dB}$, Noise Figure $\text{NF}_2 = 3\,\text{dB}$.
- **Stage 3 (Down-conversion Mixer):** Conversion gain $G_3 = -6\,\text{dB}$ (i.e. 6 dB conversion loss), Noise Figure $\text{NF}_3 = 10\,\text{dB}$.

Questions for Task 4:

- **4.1 (6.0 Credits)** Noise analysis of the cascaded receiver frontend:

  1. State the linear noise factor F_1 and linear available gain G_1 of the passive filter/feed line (Stage 1) at $T_0 = 290\,\text{K}$.
  2. Using **Friis' formula**, calculate the **total linear noise factor** $F_{\textbf{tot}}$, the **total Noise Figure** $\textbf{NF}_{\textbf{tot}}$ (in dB), and the **equivalent system noise temperature** $T_{e,\textbf{sys}}$ (in K) of the complete 3-stage receiver cascade.
  3. What would the total Noise Figure NF'_{tot} (in dB) be if the LNA (Stage 2) were placed directly at the antenna **before** the passive filter (Stage 1), i.e., sequence Stage 2 → Stage 1 → Stage 3?

- **4.2 (3.0 Credits)** Two different amplifier gain cells A and B are available as candidates for the first stage of a receiver:

  – **Amplifier A:** Linear noise factor $F_A = 1{,}6$, linear gain $G_A = 4$ (6 dB).

  - **Amplifier B:** Linear noise factor $F_B = 1{,}8$, linear gain $G_B = 10$ (10 dB). Calculate the **Noise Measure** M for both amplifiers and determine which amplifier should be placed first in a cascade to achieve the lowest overall noise figure.

- **4.3 (3.5 Credits)** Two-port noise matching:

  1. Write down the equation for the noise factor $F(\Gamma_S)$ of a linear two-port as a function of the source reflection coefficient Γ_S and list the **four noise parameters** that characterize the two-port.
  2. How are these four noise parameters measured experimentally, and what is the physical meaning of the equivalent noise resistance R_n?

- **4.4 (5.0 Credits)** Simultaneous Noise and Power Matching using **Inductive Source Degeneration (ISD)**:

  1. Why is power matching ($\Gamma'_S = \Gamma^*_{\text{in}}$) in a standard common-source FET generally different from noise matching ($\Gamma'_S = \Gamma_{S,\text{opt}}$), and why can a simple series resistor at the gate **not** be used to match the real part of the input impedance in an LNA?
  2. An InP HEMT has a transit frequency of $f_T = 300\,\text{GHz}$ and a gate-source capacitance of $C_{gs} = 80\,\text{fF}$. Its optimum noise source impedance at $f_0 = 60\,\text{GHz}$ has a real part of $\text{Re}(Z_{S,\text{opt}}) = 37{,}7\,\Omega$. Calculate the required **source degeneration inductance** L_s (in pH) to increase the real part of the transistor's input impedance $\text{Re}(Z_1) \approx 2\pi f_T L_s$ to match $\text{Re}(Z_{S,\text{opt}})$, and state how this small inductance L_s is physically realized on a mmW MMIC layout.

- **4.5 (2.5 Credits)** In broadband and millimeter-wave LNAs, a **Cascode** topology (common-source + common-gate) is frequently used instead of a single common-source stage. State **two advantages** and **two disadvantages** of the cascode topology, and explain how its high-frequency potential instability ($\text{Re}(Z_{\text{out}}) < 0$) is stabilized in the circuit schematic.

Teil II: Ausführliche Musterlösungen, Rechenwege & Bewertungsmaßstab *(Korrektur- und Punkteschema exakt nach den Vorgaben von Prof. Dr.-Ing. Ingmar Kallfass, ILH)*

---

4 Lösungen zu Part 1: Paper & Broadband GaN/InP MMIC Design (Task 1 – 40.0 Credits)

4.1 Lösung 1.1 (2.5 Credits)

- **Comparison GaN HEMT vs. InP HEMT:**
  - **Gallium Nitride (GaN) HEMT:** Wide-bandgap semiconductor ($W_g \approx 3{,}4\,\text{eV}$) with **very high breakdown voltage** ($V_{\text{br}} >$ 20-100 V), moderate cutoff frequencies ($f_T, f_{\max} > 200\,\text{GHz}$), and **very high output power density** (W/mm) and power handling capability → Primary application: **High-power amplifiers (PAs)** in transmitters. **(1.0 credit)**
  - **Indium Phosphide (InP) HEMT:** Narrow-bandgap III-V semiconductor with **extremely high electron mobility and cutoff frequencies** ($f_T > 600\,\text{GHz}, f_{\max} > 1\,\text{THz}$), **lowest RF noise figure ($\text{NF}_{\min}$)**, but **low breakdown voltage** ($V_{\text{br}} \approx 2\,\text{V}$) and low output power → Primary application: **Ultra-low-noise amplifiers (LNAs)** and sub-mmW/THz receivers. **(1.0 credit)**

- **Johnson Figure of Merit (JFoM):**

$$V_{\text{br}} \cdot f_T = \frac{E_c \cdot v_{\text{sat}}}{2\pi} = \text{const.}$$

  Within a given semiconductor technology, scaling the device for higher speed ($f_T \uparrow$ via shorter gate length $L_g \downarrow$) inevitably reduces the breakdown voltage ($V_{\text{br}} \downarrow$). **(0.5 credits)**

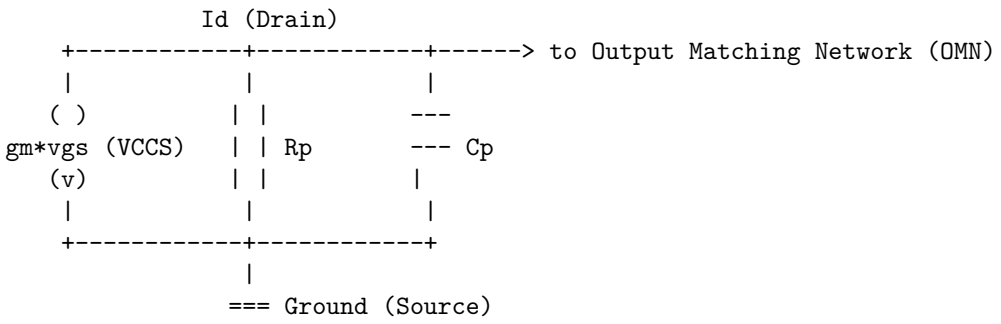
4.2 Lösung 1.2 (1.5 Credits)

- **Why SiC (or Si) substrate for GaN PAs:** Because Silicon Carbide (SiC) has a **very high thermal conductivity** ($\approx$ 3,5-4,9 $\text{W}/(\text{cm} \cdot \text{K})$), which is essential to dissipate the heat generated by the high power densities of GaN transistors. **(0.75 credits)**

- **Problem of high-voltage operation for wideband design:** High supply voltage $V_{\mathrm{D,supply}}$ leads to a high parallel load resistance $R_p \propto (V_{\mathrm{D,supply}} - V_{\mathrm{knee}})/I_{\max}$ and thus a **high-Q output equivalent circuit** ($Q = \omega R_p C_p$). According to the **Bode-Fano criterion**, a high-Q load severely limits the maximum achievable matching bandwidth. **(0.75 credits)**

4.3 Lösung 1.3 (3.0 Credits)

- **Norton-Equivalent Circuit of a Microwave Power Device:**



- **Ideal voltage-controlled current source (VCCS, $i_d = g_m v_{gs}$) (0.75 credits)**
- **Parallel output resistance** $R_p = 1/G_p$ (in shunt) **(0.75 credits)**
- **Parallel output capacitance** C_p (in shunt) **(0.75 credits)**
- **Experimental measurement method: Load-pull measurements** (varying the load impedance with impedance tuners at large-signal drive). **(0.75 credits)**

4.4 Lösung 1.4 (6.5 Credits)

- **Given values:**
  - $I_{\max} = 4\,\mathrm{A}$
  - $V_{\mathrm{knee}} = 200\,\mathrm{mV} = 0{,}2\,\mathrm{V}$
  - $V_{\mathrm{D,supply}} = 28\,\mathrm{V}$
  - $C_p = 1{,}6\,\mathrm{pF} = 1{,}6 \cdot 10^{-12}\,\mathrm{F}$
  - $\beta = 1{,}5$
  - $\mathrm{RL_{dB}} = 15\,\mathrm{dB}$
- **Calculation using the Campbell Bode-Fano equation:**

$$\mathrm{BW} < \frac{4{,}343 \cdot I_{\max}}{\mathrm{RL_{dB}} \cdot C_p \cdot \beta \cdot (V_{\mathrm{D,supply}} - V_{\mathrm{knee}})} = \frac{4{,}343 \cdot 4\,\mathrm{A}}{15 \cdot 1{,}6 \cdot 10^{-12}\,\mathrm{F} \cdot 1{,}5 \cdot (28\,\mathrm{V} - 0{,}2\,\mathrm{V})}$$

$$\mathrm{BW} < \frac{17{,}372}{36 \cdot 10^{-12} \cdot 27{,}8}\,\mathrm{Hz} = \frac{17{,}372}{1{,}0008 \cdot 10^{-9}}\,\mathrm{Hz} = 17{,}358 \cdot 10^{9}\,\mathrm{Hz} \approx \mathbf{17{,}36}\,\mathrm{GHz}$$

- **Grading scheme:** *0.5 credits for each correct value in the equation (*$7 \times 0{,}5 = 3{,}5$ *credits),* $3{,}0$ *credits for the correct result with unit GHz (or Hz),* $-0{,}25$ *credits deduction for missing or wrong unit.*

4.5 Lösung 1.5 (6.0 Credits)

1. **Normalized and absolute Norton equivalent resistance R_p:**
   - **Normalized resistance $R_{p,\mathbf{norm}}$ (for $I_{\max,\mathbf{norm}} = \frac{4\,\mathbf{A}}{5\,\mathbf{mm}} = 0{,}8\,\mathbf{A/mm}$):**

$$R_{p,\mathrm{norm}} = \frac{\beta \cdot (V_{\mathrm{D,supply}} - V_{\mathrm{knee}})}{I_{\max,\mathrm{norm}}} = \frac{1{,}5 \cdot (28\,\mathrm{V} - 0{,}2\,\mathrm{V})}{0{,}8\,\mathrm{A/mm}} = \frac{41{,}7\,\mathrm{V}}{0{,}8\,\mathrm{A/mm}} = \mathbf{52{,}125\,\Omega} \cdot \mathrm{mm}$$

(0.5 credits per value in equation = 2.0 credits, +1.5 credits for result with unit $\Omega \cdot mm$*,* $-0{,}25$ *for wrong unit).*

- **Absolute resistance R_p (for $W_{\mathbf{tot}} = 5\,\mathbf{mm}$ bzw. $I_{\max} = 4\,\mathbf{A}$):**

$$R_p = \frac{R_{p,\text{norm}}}{W_{\text{tot}}} = \frac{\beta \cdot (V_{\text{D,supply}} - V_{\text{knee}})}{I_{\max}} = \frac{1{,}5 \cdot 27{,}8\,\text{V}}{4\,\text{A}} = \mathbf{10{,}425\,\Omega}$$

(1.0 credit incl. unit Ω).

2. **Maximum linear Class-A output power $P_{\mathbf{out,max}}$ (Cripps load line, $\beta = 1$):**

$$P_{\text{out,max}} = \frac{1}{4} I_{\max}(V_{\text{D,supply}} - V_{\text{knee}}) = \frac{1}{8} I_{\max}(V_{\max} - V_{\text{knee}}) = \frac{1}{4} \cdot 4\,\text{A} \cdot (28\,\text{V} - 0{,}2\,\text{V}) = \mathbf{27{,}8}\,\text{W}$$

$$P_{\text{out,max,dBm}} = 10 \log_{10}\left(\frac{27.800\,\text{mW}}{1\,\text{mW}}\right) = \mathbf{44{,}44}\,\text{dBm} \quad (14{,}44\,\text{dBW})$$

(1.5 credits for 27,8 W and 44,44 dBm, −0,25 for wrong unit).

4.6 Lösung 1.6 (2.0 Credits)

- **Option 1: Lower the drain supply voltage $V_{\mathbf{D,supply}}$** (reduces R_p and the load quality factor Q, increasing BW). **Trade-off:** Directly reduces the maximum achievable RF output power $P_{\text{out,max}}$! **(1.0 credit)**
- **Option 2: Accept a worse input/output reflection coefficient (lower Return Loss $\mathbf{RL_{dB}}$, e.g. 10 dB instead of 15 dB)**. **Trade-off:** Higher mismatch loss and standing waves at the ports. *(Alternative valid answer: Intentional resistive loading / lossy matching, which trades off gain and efficiency for bandwidth).* **(1.0 credit)**

4.7 Lösung 1.7 (2.0 Credits)

- **Fundamental idea of distributed amplification (TWA):** Instead of trying to tune out the transistor's input and output capacitances ($C_{\text{in}}, C_{\text{out}}$) with resonant reactive matching networks (which is limited by Bode-Fano), the capacitances of multiple smaller transistors are **absorbed as distributed shunt capacitances into high-impedance inductive transmission line sections** ($Z'_L > 50\,\Omega$), effectively compensating C_p over a huge frequency range up to the cutoff frequency of the line. **(1.0 credit)**
- **Crucial passive components: Artificial transmission lines** (künstliche Übertragungsleitungen: artificial gate line and artificial drain line). **(1.0 credit)**

4.8 Lösung 1.8 (4.5 Credits)

1. **Characteristic impedance of the artificial transmission lines ($C_{\mathbf{tot,sec}} = C_{\mathbf{sec}} + C_{\mathbf{in}} = 20\,\mathbf{fF} + 120\,\mathbf{fF} = 140\,\mathbf{fF}$):**

$$Z_{L,g} = Z_{L,d} = \sqrt{\frac{L'}{C' + C_{\text{in}}/l}} = \sqrt{\frac{L_{\text{sec}}}{C_{\text{sec}} + C_{\text{in}}}} = \sqrt{\frac{350 \cdot 10^{-12}\,\text{H}}{(20 + 120) \cdot 10^{-15}\,\text{F}}} = \sqrt{\frac{350.000}{140}}\,\Omega = \sqrt{2500}\,\Omega = \mathbf{50\,\Omega}$$

(2.0 credits incl. unit Ω).

2. **Bragg cutoff frequency f_c:**

$$f_c = \frac{1}{\pi\sqrt{L_{\text{sec}}(C_{\text{sec}} + C_{\text{in}})}} = \frac{1}{\pi\sqrt{350 \cdot 10^{-12}\,\text{H} \cdot 140 \cdot 10^{-15}\,\text{F}}} = \frac{1}{\pi \cdot 7 \cdot 10^{-12}\,\text{s}} \approx \mathbf{45{,}47}\,\text{GHz}$$

(1.5 credits incl. unit GHz).

3. **Phase synchronization condition for constructive output wave interference:**

$$\beta_g \cdot l_g = \beta_d \cdot l_d$$

(The electrical phase delay per section on the artificial gate line must equal the phase delay per section on the artificial drain line so that the forward-traveling drain currents add up in phase). **(1.0 credit)**

4.9 Lösung 1.9 (3.0 Credits)

- **Comparison:** The **input capacitance** C_{in} ($\approx C_{gs}$) of a transistor is significantly **higher** than its output capacitance C_{out} ($\approx C_{ds}$). **(1.0 credit)**
- **Consequence for TWA:** Since $f_{c,g} = \frac{1}{\pi\sqrt{L_g(C_g+C_{\text{in}})}}$, the **artificial gate line has a much lower cutoff frequency and limits the overall bandwidth** of the distributed amplifier (and makes phase velocity equalization $\beta_g l_g = \beta_d l_d$ for $Z_{0g} = Z_{0d} = 50\,\Omega$ difficult). **(1.0 credit)**
- **Solution:** Insert **additional series capacitors** $C_{g,k}$ **in front of each transistor gate** (capacitive voltage divider at the input), which reduces the effective capacitance loading the gate line to $C_{\text{in,eff}} = \frac{C_g C_{\text{in}}}{C_g+C_{\text{in}}} \approx C_{\text{out}}$. **(1.0 credit)**

4.10 Lösung 1.10 (3.0 Credits)

- **Three reasons why a uniform distributed amplifier has reduced performance as a PA:**
  1. **Power dissipation in the reverse drain-line termination ("termination dump"):** Up to half of the generated RF output power travels backwards to the left drain termination resistor ($50\,\Omega$) and is wasted as heat. **(1.0 credit)**
  2. **RF drive signal attenuation along the gate line:** Due to gate-line losses ($\text{Re}(Y_{\text{in}}) \propto \omega^2$), the RF input voltage decreases moving down the gate line, so the first transistor saturates early while later transistors are under-driven. **(1.0 credit)**
  3. **Non-optimal and varying load impedance (transistor loading):** In a uniform drain line, the active load impedance seen by each transistor varies from stage to stage and over frequency instead of presenting the optimal Cripps load R_p to every transistor. **(1.0 credit)**

4.11 Lösung 1.11 (4.0 Credits)

- **Circuit topology: Nonuniform Distributed Power Amplifier (NDPA). (1.0 credit)**
- **Tapering of series gate capacitances ($C_{g1} < C_{g2} < \cdots < C_{gn}$):** Use the **smallest series gate capacitance near the input** (stage 1) and progressively larger capacitances towards the end of the gate line, so that each transistor receives an equal share of the RF input power despite gate-line attenuation. **(1.0 credit)**
- **Tapering of drain transmission line characteristic impedance ($Z_{0,d1} > Z_{0,d2} > \cdots > Z_{0,dn}$):** Use the **highest characteristic impedance near the input** (first stage, $Z_{0,d1} = R_p$) and step-wise **decrease the characteristic impedance towards the output** ($Z_{0,d,k} = R_p/k$), eliminating the reverse drain termination resistor and ensuring optimal load-line matching for all stages. **(1.0 credit)**
- **Geometrical layout parameter:** The **conductor line width** w (narrower line ⇒ higher Z_0; wider line ⇒ lower Z_0). **(1.0 credit)**

4.12 Lösung 1.12 (2.0 Credits)

- **Two transformer configurations discussed in the paper: Ruthroff transformer (0.75 credits)** and **Trifilar transformer (0.75 credits)**.
- **Additional option used in Figure 13 of the paper: (Multi-section) Quarter-wave ($\lambda/4$) transmission line transformer. (0.5 credits)**

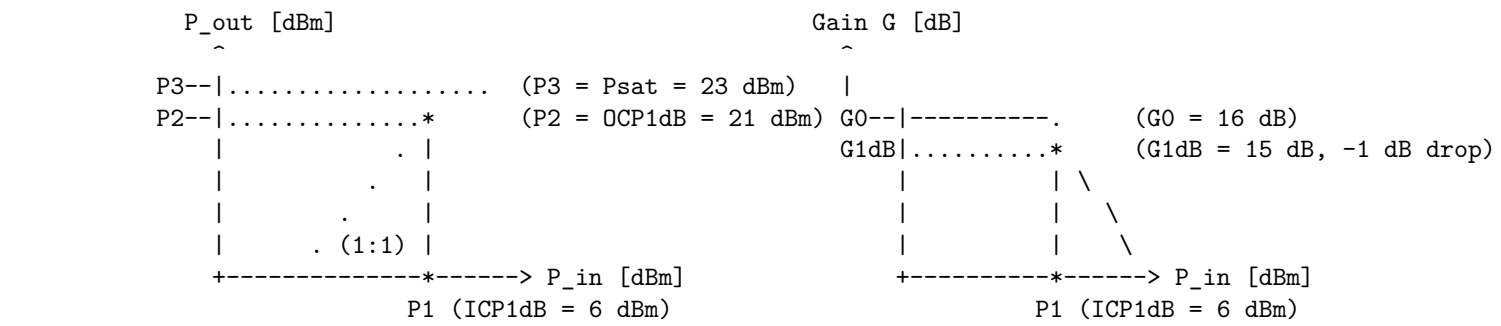
---

5 Lösungen zu Part 2: General Questions (Tasks 2, 3, 4 – 60.0 Credits)

5.1 Lösungen zu Task 2: Power Amplifiers, Nonlinearity & Dynamic Range (20.0 Credits)

Lösung 2.1 (2.5 Credits)

- **Qualitative plots of $P_{\text{out}}(P_{\text{in}})$ and $G(P_{\text{in}})$ for a one-tone sweep $u_{\text{in}}(t) = A\cos(\omega_0 t)$:**



- *0.5 credits each for correctly marking and labeling $P_1 = ICP_{1dB}$, $P_2 = OCP_{1dB}$, $P_3 = P_{sat}$, G_0 (small-signal gain), and $G_{1dB} = G_0 - 1\,dB$ ($5 \times 0{,}5 = 2{,}5$ credits).*

Lösung 2.2 (4.5 Credits)

1. **Output-related 1 dB compression point $\mathbf{OCP_{1dB}}$:**

$$G_{1\mathrm{dB}} = G_0 - 1\,\mathrm{dB} = 16\,\mathrm{dB} - 1\,\mathrm{dB} = 15\,\mathrm{dB}$$

$$\mathrm{OCP}_{1\mathrm{dB}} = \mathrm{ICP}_{1\mathrm{dB}} + G_{1\mathrm{dB}} = 6\,\mathrm{dBm} + 15\,\mathrm{dB} = +\mathbf{21}\,\mathrm{dBm}$$

$$P_{\mathrm{out,1dB}} = 10^{\frac{21}{10}}\,\mathrm{mW} \approx \mathbf{125{,}89}\,\mathrm{mW}, \qquad P_{\mathrm{in,1dB}} = 10^{\frac{6}{10}}\,\mathrm{mW} \approx \mathbf{3{,}98}\,\mathrm{mW}$$

(1.5 credits incl. units dBm and mW).

2. **DC power consumption $P_{\mathbf{DC}}$:**

$$P_{\mathrm{DC}} = V_{DD} \cdot I_{\mathrm{DC}} = 5\,\mathrm{V} \cdot 120\,\mathrm{mA} = \mathbf{600}\,\mathrm{mW} = 0{,}6\,\mathrm{W}$$

(1.0 credit incl. unit mW).

3. **Drain efficiency η_D and Power-Added Efficiency (PAE) at $\mathbf{P_{1dB}}$:**

$$\eta_D = \frac{P_{\mathrm{RF,out}}}{P_{\mathrm{DC}}} = \frac{125{,}89\,\mathrm{mW}}{600\,\mathrm{mW}} = 0{,}2098 \approx \mathbf{21{,}0}\,\%$$

$$\mathrm{PAE} = \frac{P_{\mathrm{RF,out}} - P_{\mathrm{RF,in}}}{P_{\mathrm{DC}}} = \eta_D \left(1 - \frac{1}{G_{\mathrm{1dB,lin}}}\right) = \frac{125{,}89\,\mathrm{mW} - 3{,}98\,\mathrm{mW}}{600\,\mathrm{mW}} = \frac{121{,}91\,\mathrm{mW}}{600\,\mathrm{mW}} = 0{,}2032 \approx \mathbf{20{,}3}\,\%$$

(2.0 credits: 1.0 credit for η_D, 1.0 credit for PAE, $-0{,}25$ for missing unit).

Lösung 2.3 (2.0 Credits)

- **Communication amplifiers (OFDM/QAM): $\mathbf{OCP_{1dB}}$ ($P_{\mathbf{1dB}}$)** is more relevant, because modulated communication signals have a varying envelope (high PAPR) and require high **linearity** (operation in back-off from $P_{1\mathrm{dB}}$) to prevent amplitude/phase distortion and spectral regrowth / intermodulation. **(1.0 credit)**
- **Radar amplifiers (FMCW): $P_{\mathbf{sat}}$** is more relevant, because FMCW radar uses constant-envelope frequency-modulated signals where the PA can be driven deep into saturation to maximize output power (maximum radar detection range) and efficiency (PAE). **(1.0 credit)**

Lösung 2.4 (2.0 Credits)

- **DC operating point shift (Self-Biasing):** Caused by the **second-order nonlinearity** $a_2 = \frac{1}{2}\frac{d^2I}{dV^2}$, since $\sin^2(\omega_0 t) = \frac{1}{2}(1 - \cos(2\omega_0 t))$ produces a DC current shift $\Delta I_{\mathrm{DC}} = \frac{1}{2}a_2 v_1^2 = \frac{v_1^2}{4}\frac{d^2I}{dV^2}$, which scales with the **square of the input voltage amplitude** ($v_1^2 \propto P_{\mathbf{in}}$). **(1.0 credit)**
- **Gain compression at ω_0:** Caused by the **third-order nonlinearity** $a_3 = \frac{1}{6}\frac{d^3I}{dV^3} < 0$, since $\sin^3(\omega_0 t) = \frac{1}{4}(3\sin(\omega_0 t) - \sin(3\omega_0 t))$ produces a fundamental component $\Delta I_{\omega_0} = \frac{3}{4}a_3 v_1^3 = \frac{v_1^3}{8}\frac{d^3I}{dV^3}$, which scales with the **cube of the input amplitude** (v_1^3). **(1.0 credit)**

Lösung 2.5 (3.0 Credits)

1. **Exact IM_3 frequencies ($\Delta f = f_2 - f_1 = 0{,}01\,\mathbf{GHz} = 10\,\mathbf{MHz}$):**

$$f_{\mathrm{IM3,low}} = 2f_1 - f_2 = 2 \cdot 77{,}00\,\mathrm{GHz} - 77{,}01\,\mathrm{GHz} = \mathbf{76{,}99}\,\mathrm{GHz}$$

$$f_{\mathrm{IM3,high}} = 2f_2 - f_1 = 2 \cdot 77{,}01\,\mathrm{GHz} - 77{,}00\,\mathrm{GHz} = \mathbf{77{,}02}\,\mathrm{GHz}$$

   Why most harmful: Unlike harmonics ($2f_1 = 154\,\mathrm{GHz}, 3f_1 = 231\,\mathrm{GHz}$), which lie far outside the passband and are easily suppressed by the output matching network/filter, the IM_3 products fall **directly into the operating band (in-band, close to the carrier)** and **cannot be filtered out**! **(2.0 credits)**

2. **Slope of IM_3 power:** Since $|I_{\mathrm{IM3}}| \propto v_1^3$, $P_{\mathrm{IM3}} \propto P_{\mathrm{in}}^3$, meaning the IM_3 power rises with a **slope of** $3:1$ **in dB/dB** (or 9 dB/octave), i.e., 3 dB increase for every 1 dB increase in input power. **(1.0 credit)**

Lösung 2.6 (4.0 Credits)

1. **Calculation of OIP_3 and IIP_3:**

$$\mathrm{OIP_3} = P_{\mathrm{out}} + \frac{\Delta P}{2} = +10\,\mathrm{dBm} + \frac{40\,\mathrm{dBc}}{2} = +\mathbf{30}\,\mathrm{dBm} \quad (1\,\mathrm{W})$$

$$\mathrm{IIP_3} = \mathrm{OIP_3} - G_0 = +30\,\mathrm{dBm} - 16\,\mathrm{dB} = +\mathbf{14}\,\mathrm{dBm}$$

   (1.5 credits incl. units dBm).

2. **Calculation of MDS_{in}, MDS_{out}, and SFDR (for $B = 100\,\mathbf{MHz} = 10^8\,\mathbf{Hz} \implies 10\log_{10}(B) = 80\,\mathbf{dBHz}$):**

$$\mathrm{MDS_{in}} = kT_0B + 3\,\mathrm{dB} + \mathrm{NF} = -174\,\mathrm{dBm/Hz} + 80\,\mathrm{dBHz} + 3\,\mathrm{dB} + 7\,\mathrm{dB} = -\mathbf{84}\,\mathrm{dBm}$$

$$\mathrm{MDS_{out}} = \mathrm{MDS_{in}} + G_0 = -84\,\mathrm{dBm} + 16\,\mathrm{dB} = -\mathbf{68}\,\mathrm{dBm}$$

$$\mathrm{SFDR} = \mathrm{DR}_f = \frac{2}{3}\left(\mathrm{OIP_3} - \mathrm{MDS_{out}}\right) = \frac{2}{3}\left(30\,\mathrm{dBm} - (-68\,\mathrm{dBm})\right) = \frac{2}{3} \cdot 98\,\mathrm{dB} = \mathbf{65{,}33}\,\mathrm{dB}$$

   (2.5 credits: 1.0 credit for MDS_{in}, 0.5 credits for MDS_{out}, 1.0 credit for SFDR incl. units).

Lösung 2.7 (2.0 Credits)

- **Class A vs. Class B:**
  - **Class A:** Conduction angle $\theta_c = 2\pi = 360°$ (conducts over full cycle), maximum theoretical drain efficiency $\eta_{\max} = \mathbf{50}\,\%$. **(0.5 credits)**
  - **Class B:** Conduction angle $\theta_c = \pi = 180°$ (conducts over half cycle), maximum theoretical drain efficiency $\eta_{\max} = \frac{\pi}{4} \approx \mathbf{78{,}5}\,\%$. **(0.5 credits)**
- **Purpose of odd-mode suppression resistors in parallel PA stages:** Parallel transistor loops can form an unstable ring oscillator where adjacent branches oscillate 180° out of phase (**odd-mode oscillation**). Shunt resistors placed between symmetrical nodes of the parallel branches carry **zero current in regular in-phase (even-mode) operation** (hence causing no RF power loss), but dissipate energy and **suppress any odd-mode loop instability**. **(1.0 credit)**

5.2 Lösungen zu Task 3: Two-Port Stability, Smith-Chart Matching & Planar Transmission Lines (20.0 Credits)

Lösung 3.1 (5.0 Credits)

1. **Determinant Δ and Rollett stability factor k:**

$$\Delta = S_{11}S_{22} - S_{12}S_{21} = (-j0{,}6) \cdot (-j0{,}5) - (0{,}1) \cdot (j2{,}5) = -0{,}30 - j0{,}25$$

$$|\Delta|^2 = (-0{,}30)^2 + (-0{,}25)^2 = 0{,}09 + 0{,}0625 = \mathbf{0{,}1525} \quad \implies \quad |\Delta| = \sqrt{0{,}1525} \approx \mathbf{0{,}3905}$$

$$k = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} = \frac{1 - 0{,}6^2 - 0{,}5^2 + 0{,}1525}{2 \cdot 0{,}1 \cdot 2{,}5} = \frac{1 - 0{,}36 - 0{,}25 + 0{,}1525}{0{,}5} = \frac{0{,}5425}{0{,}5} = \mathbf{1{,}085}$$

   (2.0 credits: 1.0 credit for Δ and $|\Delta|$, 1.0 credit for $k = 1{,}085$).

2. **Unconditional stability check:**
   - Since $k = 1{,}085 > 1$ AND $|\Delta| = 0{,}3905 < 1$ (as well as $|S_{11}| < 1$ and $|S_{22}| < 1$), the transistor is **unconditionally stable** at $f_0 = 60\,\text{GHz}$ for all passive source and load terminations. **(1.0 credit)**
3. **Calculation of MSG and MAG:**

$$\text{MSG} = \frac{|S_{21}|}{|S_{12}|} = \frac{2{,}5}{0{,}1} = 25 \quad \Longrightarrow \quad \text{MSG}_{\text{dB}} = 10\log_{10}(25) \approx \mathbf{13{,}98}\,\text{dB}$$

Since $k > 1$, the **Maximum Available Gain (MAG)** exists and equals:

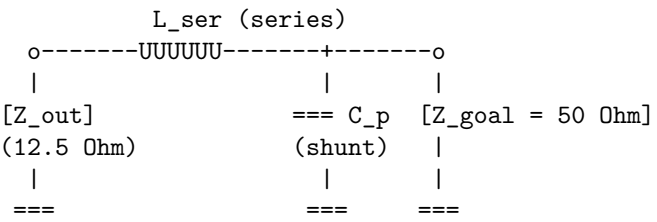
$$\text{MAG} = \frac{|S_{21}|}{|S_{12}|}\left(k - \sqrt{k^2-1}\right) = 25 \cdot \left(1{,}085 - \sqrt{1{,}085^2 - 1}\right) = 25 \cdot (1{,}085 - 0{,}421) = 25 \cdot 0{,}664 = 16{,}60$$

$$\text{MAG}_{\text{dB}} = 10\log_{10}(16{,}60) \approx \mathbf{12{,}20}\,\text{dB}$$

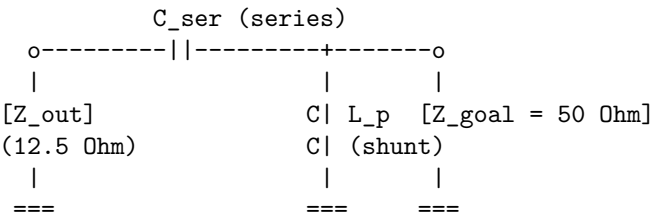
(2.0 credits: 1.0 credit for MSG = 13,98 dB, 1.0 credit for MAG = 12,20 dB).

Lösung 3.2 (4.0 Credits)

1. **Two LC L-network topologies to transform $Z_{\text{out}} = 12{,}5\,\Omega$ ($z_{\text{out}} = 0{,}25$) to $Z_{\text{goal}} = 50\,\Omega$ ($z_{\text{goal}} = 1{,}0$):**
   - **Topology A (Low-Pass L-Network):** First a **series inductor** L_{ser} connected to Z_{out}, followed by a **shunt (parallel) capacitor** C_p to ground across Z_{goal}:



   - **Topology B (High-Pass L-Network):** First a **series capacitor** C_{ser} connected to Z_{out}, followed by a **shunt (parallel) inductor** L_p to ground across Z_{goal}:



(2.0 credits: 1.0 credit per schematic with correct series/shunt designation).

1. **Transformation paths in the Smith chart:**
   - **Path A ($L_{\text{ser}} + C_p$):** Starting at $z_{\text{out}} = 0{,}25$ on the real axis, L_{ser} moves **clockwise (upwards into the inductive upper half-plane) along the constant resistance circle** $r = 0{,}25$ until intersecting the constant conductance circle $g = 1{,}0$ (at $z_A = 0{,}25 + j0{,}433$). From there, C_p moves **clockwise (downwards) along the constant conductance circle** $g = 1{,}0$ into the center $z_{\text{goal}} = 1{,}0$. **(1.0 credit)**
   - **Path B ($C_{\text{ser}} + L_p$):** Starting at $z_{\text{out}} = 0{,}25$, C_{ser} moves **counter-clockwise (downwards into the capacitive lower half-plane) along the constant resistance circle** $r = 0{,}25$ until intersecting the constant conductance circle $g = 1{,}0$ (at $z_B = 0{,}25 - j0{,}433$). From there, L_p moves **counter-clockwise (upwards) along the constant conductance circle** $g = 1{,}0$ into the center $z_{\text{goal}} = 1{,}0$. **(1.0 credit)**

Lösung 3.3 (3.0 Credits)

- **Bandwidth comparison:** Both networks achieve **the exact same matching bandwidth** (by symmetry of their transformation paths about the real axis, reaching the same maximum nodal quality factor $Q_n = \sqrt{\frac{50}{12{,}5} - 1} = \sqrt{3} \approx 1{,}73$). **(1.0 credit)**
- **Relevant criterion:** The **Bode-Fano criterion** (or Fano bandwidth limitation). **(1.0 credit)**
- **Graphical representation in the Smith chart: Constant quality factor contours (Constant-Q circles/arcs)** defined by $Q_n = \frac{|\text{Im}(Z)|}{\text{Re}(Z)} = \frac{|\text{Im}(Y)|}{\text{Re}(Y)} = \text{const.}$ **(1.0 credit)**

Lösung 3.4 (4.0 Credits)

1. **Millimeter-wave distributed replacements for lumped elements:**
   - **Shunt capacitor C_p:** Replaced by an **open-circuited stub (Open Stub)** of length $l < \lambda'/4$ (or a radial stub). **(0.5 credits)**
   - **Shunt inductor L_p:** Replaced by a **short-circuited stub (Shorted Stub)** of length $l < \lambda'/4$ (also ideal for feeding DC bias!). **(0.5 credits)**
   - **Series inductor $L_{\mathbf{ser}}$:** Replaced by a short section of **(high-impedance) transmission line**. **(0.5 credits)**
2. **Single-element matching between $Z_{\mathbf{out}} = 12{,}5\,\Omega$ and $Z_{\mathbf{goal}} = 50\,\Omega$:**
   - Use a **Quarter-Wave Transformer ($\lambda'/4$ line)**. **(0.5 credits)**
   - **Required characteristic impedance Z_T:**

$$Z_T = \sqrt{Z_{\text{out}} \cdot Z_{\text{goal}}} = \sqrt{12{,}5\,\Omega \cdot 50\,\Omega} = \sqrt{625}\,\Omega = \mathbf{25\,\Omega}$$

   (1.0 credit incl. unit Ω).
   - **Guided wavelength λ' and physical length $l = \lambda'/4$ at $f_0 = 60\,\mathbf{GHz}$ ($\sqrt{\varepsilon_{\mathbf{eff}}} = \sqrt{7{,}84} = 2{,}8$):**

$$\lambda' = \frac{c_0}{f_0\sqrt{\varepsilon_{\text{eff}}}} = \frac{3 \cdot 10^8\,\text{m/s}}{60 \cdot 10^9\,\text{Hz} \cdot 2{,}8} = \frac{5\,\text{mm}}{2{,}8} \approx 1{,}7857\,\text{mm} = 1785{,}7\,\mu\text{m}$$

$$l = \frac{\lambda'}{4} = \frac{1785{,}7\,\mu\text{m}}{4} \approx \mathbf{446{,}4}\,\mu\text{m} \quad (0{,}446\,\text{mm})$$

   (1.0 credit incl. unit μm or mm).

Lösung 3.5 (4.0 Credits)

1. **Advantages & Disadvantages of CPW vs. MSL:**
   - **Advantages of CPW (any two):** (a) No backside metallization or through-substrate via-holes required (low-inductance ground right next to transistor sources); (b) Direct compatibility with coplanar GSG on-wafer RF probes; (c) Line dimensions (w and gap s) can be scaled to match device/pad geometry while keeping $Z_0 = f(s, w)$ constant; (d) Lower cross-coupling between adjacent lines. **(1.0 credit)**
   - **Disadvantages of CPW (any two):** (a) Higher conductor attenuation/loss than MSL due to current crowding at the slot edges; (b) Larger chip area consumption due to top-side ground planes; (c) Requires airbridges to suppress parasitic slotline modes. **(1.0 credit)**
2. **Parasitic mode on CPW & suppression:**
   - The parasitic **odd mode (slotline mode)**, where the two ground planes are at different potentials ($\Phi_1 \neq \Phi_2$). **(0.5 credits)**
   - Suppressed by repeatedly short-circuiting the two ground planes across the signal line using **airbridges (Luftbrücken)** on a second metal layer (or bond wires). **(0.5 credits)**
3. **Suppression of substrate modes above $100\,\mathbf{GHz}$:**
   - Suppressed by **wafer thinning**, **backside metallization**, and inserting periodic **through-substrate via-holes (TSVs)** connecting top and bottom ground planes. **(0.5 credits)**
   - Resulting waveguide structure: **Grounded Coplanar Waveguide (GCPW)**. **(0.5 credits)**

5.3 Lösungen zu Task 4: Low Noise Amplifiers (LNAs), Cascaded Noise & Simultaneous Matching (20.0 Credits)

Lösung 4.1 (6.0 Credits)

1. **Linear parameters of Stage 1 (passive filter/feed line with $L_1 = 2\,\mathbf{dB}$ at $T_0 = 290\,\mathbf{K}$):**
   - For a passive attenuator in thermal equilibrium ($T = T_0$), the noise factor equals its loss factor L_1:

$$F_1 = L_1 = 10^{\frac{2}{10}} \approx \mathbf{1{,}585}, \qquad G_1 = \frac{1}{L_1} = 10^{-\frac{2}{10}} \approx \mathbf{0{,}631}$$

   (1.5 credits).

2. **Linear parameters of Stage 2 (LNA) and Stage 3 (Mixer) & Friis cascade calculation:**
   - Stage 2 (LNA): $\mathrm{NF}_2 = 3\,\mathrm{dB} \implies F_2 = 10^{0,3} \approx 1{,}995 \approx 2{,}0$; $G_2 = 20\,\mathrm{dB} \implies G_{2,\mathrm{lin}} = 10^2 = 100$.
   - Stage 3 (Mixer): $\mathrm{NF}_3 = 10\,\mathrm{dB} \implies F_3 = 10^{1,0} = 10$; $G_3 = -6\,\mathrm{dB} \implies G_{3,\mathrm{lin}} = 10^{-0,6} \approx 0{,}251$.
   - **Friis formula for the sequence Stage 1 → Stage 2 → Stage 3:**

$$F_{\mathrm{tot}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 \cdot G_2} = 1{,}585 + \frac{1{,}995 - 1}{0{,}631} + \frac{10 - 1}{0{,}631 \cdot 100} = 1{,}585 + 1{,}577 + 0{,}143 = \mathbf{3{,}305}$$

$$\mathrm{NF}_{\mathrm{tot}} = 10 \log_{10}(3{,}305) \approx \mathbf{5{,}19}\,\mathrm{dB}$$

$$T_{e,\mathrm{sys}} = T_0 \cdot (F_{\mathrm{tot}} - 1) = 290\,\mathrm{K} \cdot (3{,}305 - 1) = 290\,\mathrm{K} \cdot 2{,}305 \approx \mathbf{668{,}5}\,\mathrm{K}$$

   (3.0 credits: 1.5 credits for $F_{tot} = 3{,}305$, 0.75 credits for $NF_{tot} = 5{,}19$ dB, 0.75 credits for $T_{e,sys} = 668{,}5$ K).

3. **Reversed order (LNA placed first before passive filter: Stage 2 → Stage 1 → Stage 3):**

$$F'_{\mathrm{tot}} = F_2 + \frac{F_1 - 1}{G_2} + \frac{F_3 - 1}{G_2 \cdot G_1} = 1{,}995 + \frac{1{,}585 - 1}{100} + \frac{10 - 1}{100 \cdot 0{,}631} = 1{,}995 + 0{,}00585 + 0{,}1426 = \mathbf{2{,}143}$$

$$\mathrm{NF}'_{\mathrm{tot}} = 10 \log_{10}(2{,}143) \approx \mathbf{3{,}31}\,\mathrm{dB}$$

   (Placing the LNA before the 2 dB passive loss improves the system noise figure by almost 2 dB!). **(1.5 credits)**

Lösung 4.2 (3.0 Credits)

- **Noise Measure M definition and calculation:**

$$M = \frac{F - 1}{1 - \frac{1}{G}}$$

$$M_A = \frac{F_A - 1}{1 - \frac{1}{G_A}} = \frac{1{,}6 - 1}{1 - \frac{1}{4}} = \frac{0{,}6}{0{,}75} = \mathbf{0{,}80}, \qquad M_B = \frac{F_B - 1}{1 - \frac{1}{G_B}} = \frac{1{,}8 - 1}{1 - \frac{1}{10}} = \frac{0{,}8}{0{,}90} \approx \mathbf{0{,}889}$$

- **Decision:** Since $M_A = 0{,}80 < M_B = 0{,}889$, **Amplifier A must be placed first** ($A \to B$) to achieve the minimum overall cascade noise factor ($F_{A\to B} = 1{,}6 + \frac{1{,}8-1}{4} = 1{,}80 < F_{B\to A} = 1{,}8 + \frac{1{,}6-1}{10} = 1{,}86$). **(3.0 credits: 1.0 for M_A, 1.0 for M_B, 1.0 for correct decision)**

Lösung 4.3 (3.5 Credits)

1. **Two-port noise equation and the 4 noise parameters:**

$$F(\Gamma_S) = F_{\min} + \frac{4 r_n |\Gamma_S - \Gamma_{S,\mathrm{opt}}|^2}{(1 - |\Gamma_S|^2)\,|1 + \Gamma_{S,\mathrm{opt}}|^2} \qquad \left(\text{with } r_n = \frac{R_n}{Z_0} \text{ or in admittance form } F = F_{\min} + \frac{R_n}{G_S}|Y_S - Y_{S,\mathrm{opt}}|^2\right)$$

   - **The 4 noise parameters:** (1) Minimum noise factor $F_{\min}$, (2) Normalized equivalent noise resistance $r_n = R_n/Z_0$, (3) Magnitude $|\Gamma_{S,\mathbf{opt}}|$ (or $G_{S,\mathrm{opt}}$), and (4) Phase $\angle\Gamma_{S,\mathbf{opt}}$ (or $B_{S,\mathrm{opt}}$) of the optimum source reflection coefficient. **(2.0 credits)**

2. **Measurement method & meaning of R_n:**
   - Measured by **Source-Pull measurements** (varying the source impedance Γ_S with an input tuner, measuring $F(\Gamma_S)$, and fitting the **noise paraboloid**). **(0.75 credits)**
   - **Meaning of R_n:** R_n quantifies the **sensitivity (penalty) of the noise figure to source mismatch** $\Gamma_S \neq \Gamma_{S,\mathrm{opt}}$ (a larger R_n yields a steeper noise paraboloid and smaller noise circles). **(0.75 credits)**

Lösung 4.4 (5.0 Credits)

1. **Why $\Gamma_{S,\mathbf{opt}} \neq \Gamma^*_{\mathbf{in}}$ and why no series gate resistor:**
   - The input of a FET is predominantly capacitive (C_{gs}) with a very small series resistance $\mathrm{Re}(Z_{\mathrm{in}}) = R_g + R_{gs} + R_s$, whereas the optimum noise impedance $Z_{S,\mathrm{opt}}$ has a much larger real part. Thus $\Gamma^*_{\mathrm{in}} \neq \Gamma_{S,\mathrm{opt}}$. **(1.0 credit)**
   - Adding an ohmic series resistor at the gate would introduce **thermal noise** ($\overline{v_n^2} = 4kTR$), which gets amplified by the transistor and severely degrades $\mathrm{NF}_{\min}$! **(1.0 credit)**

2. **Calculation of source degeneration inductance L_s:**

- Using the inductive source degeneration formula for the real part of the input impedance:

$$\mathrm{Re}(Z_1) \approx \omega_T L_s = 2\pi f_T L_s \stackrel{!}{=} \mathrm{Re}(Z_{S,\mathrm{opt}})$$

$$L_s = \frac{\mathrm{Re}(Z_{S,\mathrm{opt}})}{2\pi f_T} = \frac{37{,}7\,\Omega}{2\pi \cdot 300 \cdot 10^9\,\mathrm{Hz}} = \frac{37{,}7}{1{,}885 \cdot 10^{12}}\,\mathrm{H} \approx \mathbf{20{,}0 \cdot 10^{-12}\,H = 20\,pH}$$

- **MMIC realization at mmW frequencies:** Realized as a short **short-circuited transmission line stub (Shorted Stub)** between the FET source terminal and ground. **(3.0 credits: 1.0 credit for formula, 1.5 credits for $L_s = 20\,$pH incl. unit, 0.5 credits for Shorted Stub)**

Lösung 4.5 (2.5 Credits)

- **Advantages of the Cascode (any two):** (1) Strongly reduced **Miller effect** on the input CS transistor ($A_{v,\mathrm{CS}} \approx -1 \implies$ higher bandwidth); (2) Higher **gain** (MSG/MAG) and much higher **output impedance** $Z_{\mathrm{out}} \gg r_{ds}$; (3) Much higher **reverse isolation** (very small $|S_{12}|$, simplifies matching). **(1.0 credit)**
- **Disadvantages of the Cascode (any two):** (1) **Doubled DC power consumption** (or doubled supply voltage $V_{DD} = 2V_{DS} \implies$ lower efficiency); (2) Prone to **high-frequency instability** because the output conductance $\mathrm{Re}(Y_{22})$ / output resistance $\mathrm{Re}(Z_{\mathrm{out}})$ can become **negative** ($|S_{22}| > 1$). **(1.0 credit)**
- **Stabilization in schematic:** Inserting a carefully dimensioned **series R_g-C_g feedback network at the gate of the upper common-gate (CG) transistor** to ground, which eliminates the negative real part of the output impedance. **(0.5 credits)**